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Lithium ion transportation in lignin infused polyvinyl alcohol based eutectogel: A molecular dynamics framework

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ABSTRACT

Eutectogels are an arising material in the field of energy storage applications like high-energy Lithium-Ion batteries (LIBs) due to their properties like higher ionic conductivity, environment friendly and enhanced safety. However, it is important to understand the working mechanism of eutectogels at an atomistic level with the help of Molecular Dynamics (MD) simulations, in order to enhance their performance as electrolytes for LIBs. Eutectogels are formed by integrating deep eutectic solvent (DES) involving Lithium Chloride (LiCl) and ethylene glycol (EG) in the molar ratio of 1:5 with the polymeric matrix of Polyvinyl alcohol (PVA) chains considered for this study. Thereafter, lignin molecules are induced in the eutectogel system to improve the characteristics of an eutectogel. The elaborate study of the effect of lignin on the lithium ions movement through the lignin-added-eutectogel is involved in this work. The increase in hydrogen bonds among PVA polymeric chains and lignin molecules show the strength enhancement. From the Radial Distribution Function results it is derived that the lithium ions are showing enhanced interactions with chlorine ions than any other component of the system. Mean Square Displacement gives an idea about diffusivity of lithium ions through the systems and it suggests that the Li^+ ion diffusivity is improved after Lignin imposition. On the basis of findings from this work, the modified Lignin infused PVA based eutectogel can be the potential candidate for the battery and energy storage applications.

1. Introduction

Polyvinyl alcohol (PVA) has biocompatible and biodegradable characteristics with low toxicity [1]. High-strength gels can be formed from PVA through the abundance of crosslinking moieties via chemical crosslinking or through hydrogen bonding (H-bonding) interaction in physical crosslinking [2–4,63]. In the solvent, PVA chains act as gelators and depending upon the liquid components, hydrogel, ionogel, and organogels are formed from PVA [5–7]. The solvent medium as well as the functionalities of gelator, influence the evolution of the gel network and contribute to the mechanical properties and stability [8,9]. PVA-based biocompatible hydrogels are explored for biomedical applications, whereas vast literature is also available for dye adsorption and metal ion removal [10–12]. Organogels show better antidehydration, anti-freezing, and heat resistance than hydrogels [13,14]. However, all such properties, along with high conductivity, are difficult to obtain simultaneously. Ionogels and conductive hydrogels have ion conduction characteristics where ionic liquids (ILs) and aqueous electrolyte salts, respectively, are swollen in a polymer network. The flexibility, softness and durability of such soft matters lead to the development of flexible sensors [15–17]. However, the water evaporation and poor anti-freezing properties lead to the loss of sensory activity, resulting in poor performance and restricted application [18]. Having a greater amount of toxicity and higher cost make IL-supported ionogel sensors less attractive for commercialization PVA-based conductive hydrogels, organogels and ionogels suffer the similar drawbacks discussed above Lately, eutectogels made of deep eutectic solvent (DES) and polymer matrix, have been invented to overcome all the drawbacks of organogels (toxicity and low conductivity), hydrogels (cannot perform at lower as well as higher temperatures) and ionogels (high price, complex method of preparation, not biocompatible) [19]. Wide ranges of polymer matrices like PVA, poly(glycerol 1,3-diglycerolate diacrylate) are being explored as eutectogels [20]. To cater the rapidly increasing demands of electrical devices and robotics technology, eutectogels have been developed at a rapid pace since its inception [3,21–26].

The eutectogels are comprised of DES and a matrix composed of monomers or polymers [19,27,28]. The H-bond Donors (HBDs) and H-bond Acceptor (HBAs) interact with H-bonding and van der Waals

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(vdW) interactions and are combined with various molar ratios to form DES [29]. This novel gel material has basic properties corresponding to DES, such as high conductivity, low vapour pressure, wide liquid range, sustainability, ease of preparation, and cheap price [30]. Additionally, natural DESs are proven to be non-toxic [31]. The performance of eutectogel depends on its constituents. The eutectogels containing HBAs like choline chloride class and HBDs like urea, and oxalic acids face some drawbacks as low strength as well as low conductivity values [19]. In contrast, the eutectogel prepared from a DES containing a metal salt and hydroxyl moieties exhibits significantly high conductivity values along with remarkable gel strength [26]. Recently, the eutectogel has been adopted as a potential candidate to resolve the safety issues concerning Lithium-Ion batteries (LIBs) and to enhance its conductive capacity [32–34]. Choline chloride and ethylene glycol (EG) based DES medium facilitates the UV irradiated, free radical polymerization of PVA/ polyacrylic acid [35]. The resulting network depicts a self-recoverable eutectogel to be used as a flexible sensor. Freeze-thawing promoted sol-gel transition of PVA-gelatin polymers in DES creates eutectogel with good mechanical properties, high-temperature endurance and high ionic conductivity [36]. The sensing application in harsh chemical environments and thermal conditions lead to the potential use in electronic devices. The recyclability of network evolution focuses on the goal of minimization of electronic waste pollution. The solvent replacement process creates high mechanical strength bearing PVA-eutectogel [37]. The 'polymer dissociation' followed by the 'network reconstruction' is involved in obtaining network optimization that generates such strong eutectogels. Likewise, anti-freezing and high-strength eutectogel are reported using microphase separation of PVA in the aqueous DES solution [38]. The recrystallizing domain of PVA provides enhanced physical crosslinking points, resulting in excellent mechanical properties. However, when metallic salts like $ZnCl_2$ and LiCl are used as HBA, the resulting DESs create highly conductive, superior mechanical strength and frost-resistant eutectogel with PVA as gelator [33]. Additionally, lignin has been infused in the network for controllable and flexible PVA chains and thus opens the potential application in flexible sensing. The long chains of lignin bring multifunctionality that creates nonbonded interactions with the metallic and nonmetallic components of DES and PVA.

Though the infusion of lignin in eutectogel has been reported very recently, it is a popular choice of structural flexibility and network versatility for other types of gels like a hydrogel. Hence the potential of lignin in the eutectogel is elaborated in light of hydrogel. The mechanical and processability qualities of polyurethane hydrogels have vastly improved due to the incorporation of lignin [39]. Highly stretchable hydrogel can be produced as lignin behaves as a plasticizer and boosts the mobility of polymer chains [40]. Despite having a highly porous structure but less surface area than that of synthetic hydrogel, the lignin-based hydrogel is more stable at a wide range of temperatures and also has a low swelling affinity. The strength and stability of hydrogels are enhanced due to the presence of lignin. However, the elasticity of the lignin-added hydrogel is seemed to be decreasing at the elevated temperatures [41]. The double-layer hydrogel which can be used to harvest mechanical energy (additional adhesive strength of ~18.7 kPa and Young's modulus of ~101.3 kPa) is made by imposing lignin or cellulose derivatives on hydrogels specifically PVA hydrogels [42]. The hydrogel formed using PVA/Polyacrylic Acid and Lithium Chloride (LiCl) or Ligno-Sulphate (LS) results in a robust hydrogel with a mechanical strength of 1.04 MPa. This strength is achieved through covalent crosslinking, which enhances the hydrogel's flexibility and stability. As a result, this hydrogel demonstrates significant potential for use in various applications, including smartphones, touch-pens, and biomedical devices [43]. The hydrogel formed by PVA and lignin mixture shows the covalently crosslinked PVA and lignin also the intermolecular H-bonds between them enhance the properties of the hydrogel [39]. The observations are also drawn on the basis of conductivity, stretchability and transmittance.

Experimental studies of eutectogels face limitations in directly observing atomic-level interactions and molecular behaviors, which are crucial for understanding their structure and performance. These drawbacks hinder the detailed investigation of non-bonded interactions like H-bonding and vdW forces, which significantly influence the structural orientation, chain mobility, and ion transportation within the eutectogel framework. Molecular Dynamics (MD) simulation emerges as a powerful tool to overcome these experimental limitations. MD simulations provide a comprehensive atomic and molecular level perspective, enabling researchers to map detailed interaction networks and reveal transport behaviors that are challenging to observe experimentally. By utilizing MD simulations, researchers can study the impact of H-bonding and other non-bonded interactions on the eutectogel's structure and properties. The presence of multiple functional moieties in various components of eutectogel creates the possibility of numerous non-bonded interactions like H-bonding and vdW interactions, which directly affect the structural orientation of components, chain mobility and ion transportation mechanism. Hence the importance of study of eutectogel from the atomic and molecular level perspective is realized. Such investigation can reveal the detailed interaction mapping in the eutectogel framework and controlling such phenomena will enhance their performance for wider applications. These also lead to the functionalization of precursors for performance enhancement. MD simulation is a useful tool to map the detailed interaction mapping and reveal the transport behaviour at the atomistic level [64]. Analogues study of hydrogels in MD simulation reveals a prominent role of the application stage [44]. As H-bonding plays the primary interaction link between HBA and HBD for DES, the MD simulation is adequate to reveal the scenario. Additionally, the long-range and short-range interactions between the molecules are extensively studied with the help of MD simulations. The replication of microstructures of gel along with the mimicking the material behaviour through MD simulation provides vast information [45]. The transport behaviour like chain mobility, ionic conductivity and viscosity can be estimated through MD simulation [46]. The key properties like conductivity, diffusivity, H-bonding and stress−strain behaviour of PVA hydrogel are explored in MD framework [44,46,47]. A significant knowledge gap exists in the literature regarding the molecular-level understanding of PVA-based eutectogels. While MD simulations have been employed to study PVA-based gels in general, there is a notable absence of research focusing specifically on PVA-based eutectogels using all-atom MD simulations. This gap limits comprehension of the unique interactions and behaviors that characterize these materials. The present work addresses this limitation by presenting the first all-atom MD simulation study of PVA-based eutectogels.

For this study, the transport as well as structural properties of lignin infused PVA based eutectogels are explored employing MD simulation. The metal-salt based DES constituting LiCl and EG in a specific molar ratio is mixed with PVA to form the eutectogel. The role of PVA in the eutectogel system is multifaceted and crucial for the overall performance of the material. PVA primarily acts as a gelling agent, forming a robust three-dimensional network through H-bonding and physical entanglements. This network provides essential structural support, ensuring the gel maintains its shape and mechanical integrity, which is vital for various applications such as flexible electronics or wearable devices. Additionally, PVA interacts with the DES, by encapsulating and immobilizing it within the gel matrix. This interaction ensures uniform distribution of the DES and prevents phase separation, contributing to the stability and longevity of the gel. Moreover, the hydroxyl groups present in PVA chains allow it to act as a HBD, enabling interactions with other components in the system, which can influence properties like ionic conductivity and mechanical strength. While PVA primarily provides structural support, its ability to participate in H-bonding also affects the gel's viscoelastic properties, including elasticity, flexibility, and recovery from deformation. Thereafter, lignin is induced in the eutectogel system to improve the mechanical strength of a eutectogel. To

Table 1
Number of atoms in different eutectogels with initial box size.

System No.	Eutectogel system name	Number of atoms				Total number of atoms	Initial box geometry (nm^3)	Final box geometry (nm^3)	Reference
		LiCl	EG	PVA	lignin				
1	0-L	6400	160,000	20,232	0	186,632	18.6 × 13.9 × 13.9	15.6 × 11.7 × 11.7	[33]
2	1-L	6400	160,000	20,232	455	187,087	20.4 × 20.4 × 20.4	12.9 × 12.9 × 12.9	
3	2-L	6400	160,000	20,232	910	187,542	24.8 × 12.4 × 12.4	20.5 × 10.2 × 10.2	
4	3-L	6400	160,000	20,232	1365	187,997	37.2 × 12.4 × 12.4	26.8 × 8.9 × 8.9	

understand the effect of lignin on eutectogel, the lignin molar ratio is varied. The elaborate study of the effect of lignin on the lithium ions movement through the lignin-added-eutectogel is involved in this work. The involvement of H-bonding, a key non-bonded interaction parameter, has been mapped for the formation of DES and the evolution of the gel network. Further, a structural orientation of functional moiety in the gel network is derived from Radial Distribution Function (RDF) plots. The mobility of Li^+ ions through a gel matrix has been explored by mean squared displacement (MSD). These are the necessary revelations in the state-of-the-art eutectogel framework which are practically immeasurable from experimental techniques. The structural configuration and ionic mobility coupled with the effect of lignin will provide insights about the designing eutectogels with the desired characteristics and their applications.

2. Computational methodology

2.1. Details of systems

The eutectogel system has the primary component of DES and PVA chains mixed in various molar ratios. LiCl and EG act as HBA and HBD respectively in DES with a molar ratio of 1:5. This molar ratio of LiCl to EG is used in this study based on established findings in experimental literature [33]. This ratio has been shown to effectively form a DES by reaching the system's eutectic point, thus optimizing critical physicochemical properties such as viscosity and conductivity. These properties play a crucial role in the applications we are exploring, as they significantly influence the performance and stability of the DES in various contexts. The enhancement of mechanical properties and ion transportation of lithium through the gel has been modified with the infusion of lignin in varying numbers. Hence, four such eutectogel systems are prepared and the initial configurations are given in Table 1. The initial structures of HBA and HBD of DES, lignin, monomeric units are drawn in the Avogadro open-source package [48]. The monomeric PVA is converted into 80-repeat unit chains with the help of in-house programs through Python package which is discussed in literature [63]. The structural representations are illustrated in Fig. 1. The partial charges on every atom are computed through Gasteiger process [49].

The structural orientation and dynamic behaviour of eutectogel require accurate mapping of the bonded as well as non-bonded interactions through the inclusion of the Optimized Potentials for Liquid Simulations (OPLS) force field [50] whose functional form is given below.

$$V_{total} = \sum_{bonds} k_r (r - r_{eq})^2 + \sum_{angles} k_\theta (\theta - \theta_{eq})^2 + \sum_{dihedrals} k_\chi \left[1 + cos(n\chi - \delta_\chi)\right] + \sum_{improper} k_\psi [1 + cos(n\psi - \delta_\psi)] + \sum_{i=1}^{N-1} \sum_{i<j}^{N} \left\{ 4\varepsilon_{ij} \left[\left(\frac{\sigma_{ij}}{r_{ij}}\right)^{12} - \left(\frac{\sigma_{ij}}{r_{ij}}\right)^{6} \right] + \frac{q_i q_j}{4\pi\varepsilon_0 r_{ij}} \right\} \quad (1)$$

Here, total potential energy is represented by the term V_{total} which includes energies like bending energy of angles, stretching energy of bonds, tortional energy for proper and improper dihedrals as well as energy for non-bonded interactions like Coulombic interactions and vdW interactions. k_r represents the bond stretching constant whereas, standard length of bond is denoted by r_{eq} and the length of bond is denoted by r. k_θ is the constant involved in the calculation of bending energy of angles. θ_{eq} is the standard angle between the molecules. θ is the angle between the molecules. For proper dihedral related calculation, k_χ is the constant involved and $n\chi$ is the term involving$n$ andχrepresenting multiplicity and dihedral angle between two planes respectively. The phase shift is denoted as δ_χ. Similarly, k_ψ is the constant for improper dihedrals. The multiplicity is denoted as n and also the improper dihedral angle between the planes is denoted as ψ. δ_ψ represents phase shift term. The term related to non-bonded interactions includes, L-J potential parameters that are given as σ_{ij}andε_{ij}. The permittivity of vacuum is denoted as ε_0. The charges of the $i^{th}$and j^{th}atoms are represented as $q_i$and $q_j$respectively and the distance between those two atoms is represented as r_{ij}.

2.2. MD simulation framework

The MD simulation is performed in the GROMACS-2022.3 open-source package [51] involving the stages of energy minimization, equilibration, and production. The energy minimization step is carried out to reduce the system's initial potential energy. This assists the system to attain its stable configuration. For this step, the Steepest Descent algorithm is used. The required value of a convergence threshold is 1000 kJ/mol nm. With the 10^{-3} nm step size, 5×10^6 number of steps are executed. For the purpose of the equilibration and final production run, the Leap-frog integrator is utilized. Using this, each equation of motion is integrated at two fs time steps. During the equilibrium stage, the Lincs algorithm limits H-bonding [52]. The Particle-Mesh Ewald (PME) process is used to handle long-range electrostatic interactions [53]. The cutoff distance is 1.2 nm for vdW interactions. The GRID algorithm is applied 20 times a step to revise the neighbouring grid list. Thereafter the equilibration step is divided into two ensembles. The canonical ensemble is known as NVT, followed by the NPT ensemble. To maintain uniformity throughout the system also to obtain better dynamics, annealing is performed during NVT. During the annealing process, the system temperature raised from 300 K to 500 K at a rate of 0.4 K/ps followed by a cooling step from 500 K to 300 K at 0.1 K/ps rate. This method helps in relieving any stress within the system, allowing it to reach a more stable and relaxed state, which is crucial for accurately capturing the gel's structural and dynamic properties. The primary advantage of annealing is that it enables the system to overcome energy barriers, leading to a more representative and equilibrated configuration. After the annealing process, the system's temperature is maintained at 300 K for the rest of the simulation. The total time duration for NVT equilibration is 30 ns. In order to decrease the computational time, NPT step is performed for 10 ns to reduce the box size and obtain the desired density of system. Finally, the production run is carried out for 20 ns using NPT ensemble. During equilibration and production steps, temperature and pressure are maintained using a velocity rescaling thermostat, which is known as V-rescale [54] and Parrinello-Rahman barostat [55]. For each system, three simulations are conducted separately. Subsequently, the data from these runs is averaged to mitigate any potential errors arising from the simulations. For the purpose of visualize the trajectories, Visual Molecular Dynamics that is VMD

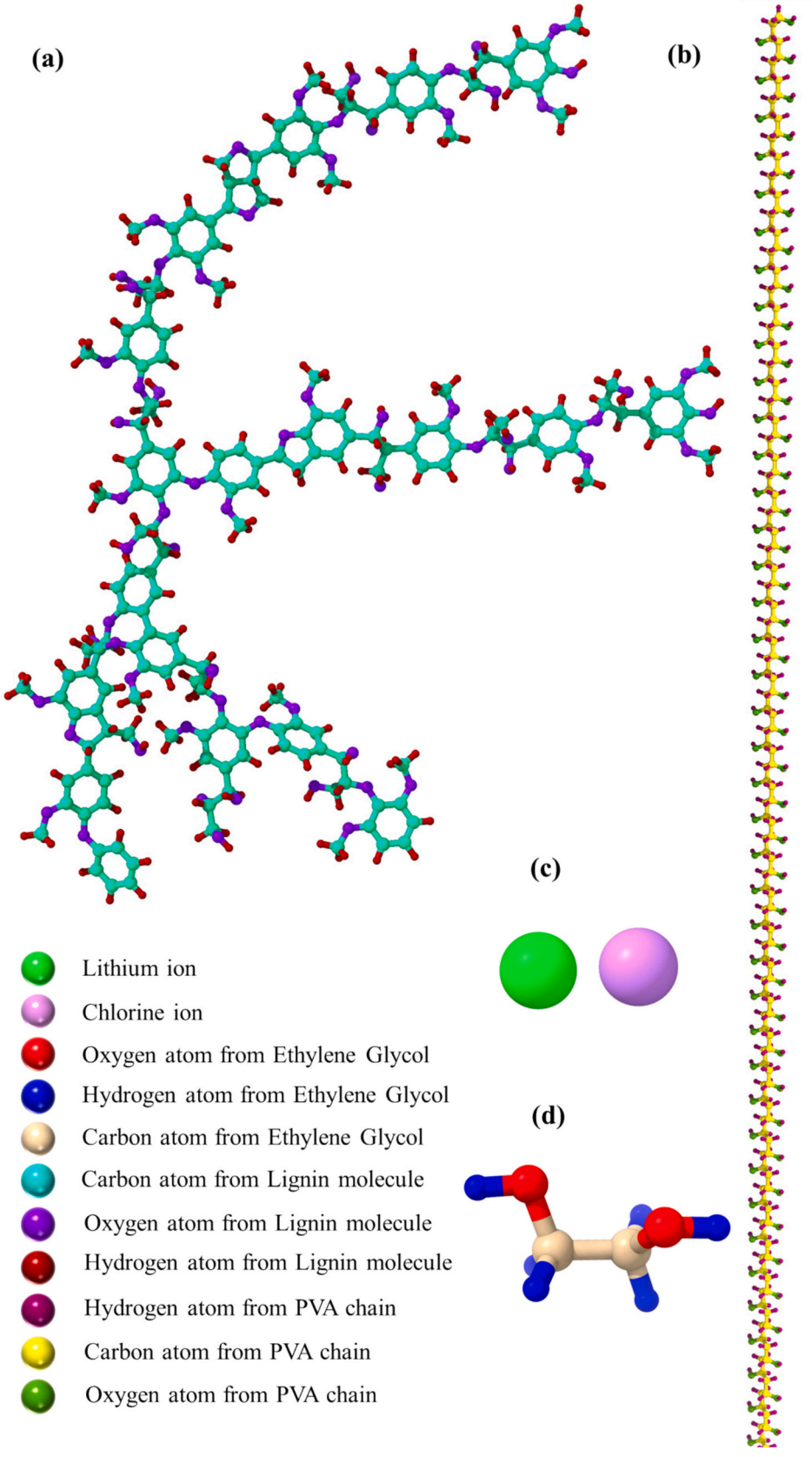


Fig. 1. Molecular structures of the components: (a) Lignin (b) PVA (c) LiCl (d) EG.

package is used [56]. For analysis trajectory, the python in-house code is used.

2.3. Analysis

2.3.1. Radial Distribution Function (RDF) and coordination number (CN)

The dispersion of different atoms present in the system with respect to one reference atom is given by the Radial Distribution Function (RDF) given in the following equation.

$$g_{PQ}(r) = \frac{\langle \rho_Q(r) \rangle}{\langle \rho_Q \rangle_{local}} = \frac{1}{\langle \rho_Q \rangle_{local}} \frac{1}{N_P} \sum_{i \in P}^{N_P} \sum_{j \in Q}^{N_Q} \frac{\delta(r_{ij} - r)}{4\pi r^2} \quad (2)$$

Here, RDF is given as $g_{PQ}(r)$ which is represented in terms of an average number density of atoms belong to Qdenoted as$\langle \rho_Q(r) \rangle$and the local average number density of atoms belong to Q and present inside the sphere of radius $dr$from the atoms belong to Pdenoted as$\langle \rho_Q \rangle_{local}$. The Dirac Delta function $\delta(r_{ij} - r)$is involved in RDF calculation where, the distance between $i^{th}$and j^{th}atoms and the central atom and another atom are represented as $r_{ij}$and $r$respectively. Total number of atoms belongs to$P$and $Q$are given by $N_P$and $N_Q$respectively. The Co-ordination number (CN) is an indicator of the bonding status. It is helpful in deriving the local arrangement as it gives the number of Q atoms surrounding to the P atom which is a reference atom. Following Eq. (3) is used to calculate the CN:

$$CN = 4\pi\rho N \int_0^r g(r) r^2 dr \quad (3)$$

Here, CN is the function of r that is the shell radius when reference atom is kept at centre. CN is calculated by integrating $g(r)$ that is RDF. The number density of particles is represented as ρ whereas number of system atoms is represented by N.

2.3.2. H-bond

The number of H-bond is calculated with the following expression.

$$C_{HB} = \frac{\langle H(\tau) H(\tau + t) \rangle}{\langle H(\tau) H(\tau) \rangle} \quad (4)$$

Here, C_{HB} is representing the quantity of H-bonds formed. The required H-bond order parameters are given as $H(\tau)$ and $H(\tau + t)$ respectively at time τ and time$(\tau + t)$.

2.3.3. Mean Square Displacement (MSD)

The Mean Square Displacement (MSD) gives the rough idea about the diffusivity of an ion in the system. Diffusion coefficient termed as D_A is calculated using Einstein relation [57] which is given by the Eq. (5).

$$\lim_{t \to \infty} \left\langle \| r_i(t) - r_i(0) \|^2 \right\rangle_{i \in P} = 6 D_A t \quad (5)$$

Here, at time 0, the position where the centre of an atom situated is given by the term $r_i(0)$as well as at time$t$, it is given as$r_i(t)$term. The term$\beta$is also an indicator of diffusivity which can be calculated as:

$$\beta = \lim_{t \to \infty} \frac{d[log(\triangle r^2)]}{d[log(t)]} \quad (6)$$

$$MSD = 6 D_A t \quad (7)$$

The value of β states the diffusion behaviour of an ion. If the value is equal to 1 the behaviour is diffusive. At β equals to 1, the Einstein relation reduces to the Eq. (7) which is known as diffusion function. Ifβvalue is >1, then the behaviour is super-diffusive which implies the fast diffusion of ions than the ideal diffusion. Ifβ value is <1, it suggests the sub-diffusive conduct because of some hindrance to the diffusion of ions.

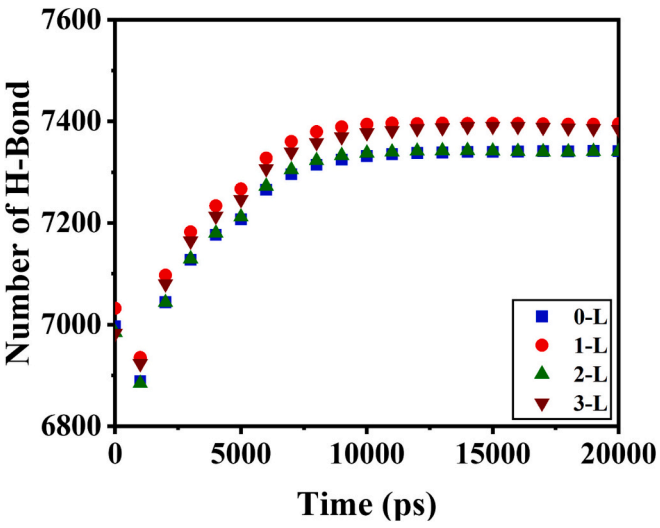


Fig. 2. Number of H-bonds between EG and EG moieties.

Table 2
Average number of H-bonds.

Sr. No.	Systems	Interactions between					
		EG-EG	Lignin-EG	Lignin-PVA	Lignin-Lignin	PVA-PVA	PVA-EG
1	0-L	7297	–	–	–	412	1016
2	1-L	7354	13	0	4	404	1013
3	2-L	7299	27	1	10	411	1005
4	3-L	7338	35	3	13	418	986

3. Results and discussion

3.1. Evolution of H-bonds

H-bonding is crucial in gel systems because it drives the formation of the network structure that gives gels their unique properties. These bonds create and maintain the gel's three-dimensional matrix, providing mechanical strength, elasticity, and the control ion mobility. Fig. 2 represents the number of H-bonds forming in due course of time between EG-EG molecules for all the systems. According to the literature, the –OH moiety can act as a H-bond donor or acceptor based on the environment and thus becomes very flexible in terms of forming the H-bonds [58]. Initially, H-bonding increases for the first 5 ns. During the equilibration phase, hydrogen atoms are constrained using the LINCS algorithm. However, after equilibration, the LINCS algorithm is not applied during the production run. Without structural constraints, the system begins to relax, leading to an initial increase in H-bonds. After this period, the number of H-bonds remains constant throughout the dynamics, indicating that the ethylene glycol has reached equilibrium. However, the average number of H-bonds is in the constant range, which gives in the Fig. S1 of supplementary materials and Table 2. This suggests the stability of the system and the lignin doesn't affect the H-bonding between EG to EG. The average numerical value of H-bonds between EG-EG corresponding to all the systems that are from 0-L to 3-L are 7297, 7354, 7299, and 7338, respectively. This indicates that EG molecules are closely associated, contributing to the overall cohesion and stability of the system. Such extensive H-bonding can also suggest that EG plays a significant role in maintaining the structural integrity and dynamics of the gel network.

The H-bond curves for Lignin-EG and Lignin-PVA are represented in Fig. 3(a) and Fig. 3(b), respectively. Fig. 3(a) represents the interaction between Lignin and EG where both are very flexible to switch between

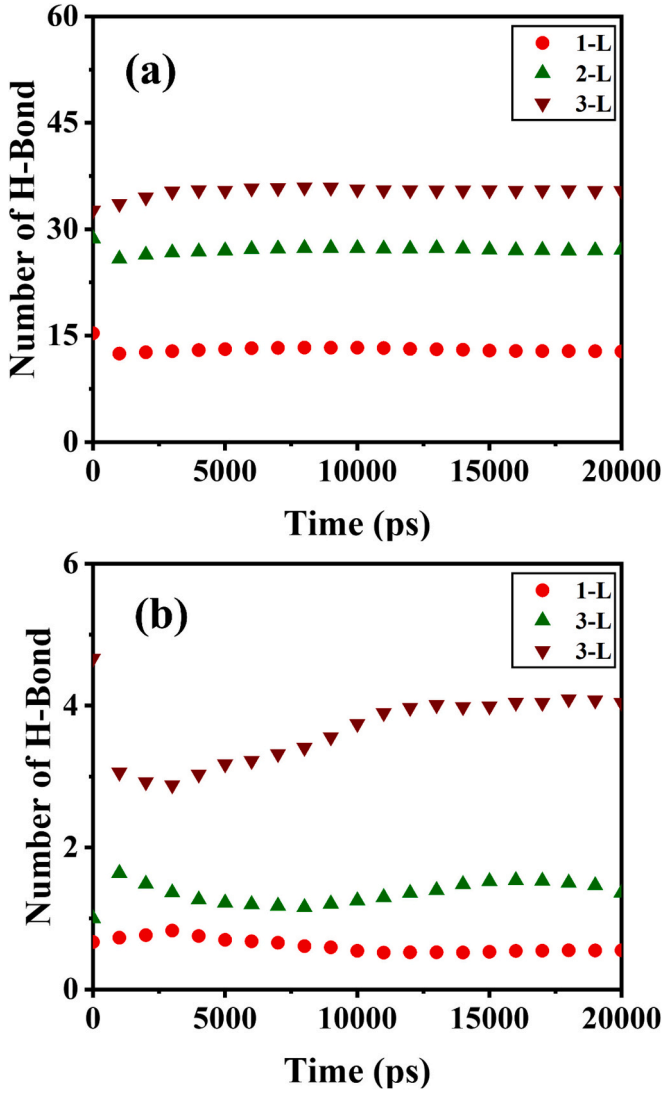


Fig. 3. Number of H-bonds between (a) lignin and EG molecules (b) lignin and PVA.

the acceptor and donor criterion. The presence of methoxy and ether moieties in lignin promote the acceptor behaviour whereas, the presence of hydroxyl moiety acts as H-bond donor [59]. The 13, 27 and 35 are the average numerical value of H-bonds for the systems involving Lignin-EG interactions that are 1-L, 2-L and 3-L, respectively. This shows the Lignin-EG interactions increases as number of lignin molecules increases in the system. This suggests that higher lignin concentrations enhance the formation of H-bonds with EG, potentially leading to a more interconnected and stable network within the system. However, LiCl is another component of DES, but it does not form H-bonds with EG. The primary H-bonding elements are nitrogen, oxygen, and fluorine. These elements are typically responsible for the formation of H-bonds in such systems. However, LiCl does not contain any of these elements, and as a result, it does not directly participate in the H-bonding interactions. The nonbonded interaction between Lignin and PVA is observed with the help of number of H-bonds curve Fig. 3(b) and. The average numerical value of H-bonds among lignin molecules to PVA is observed for the systems 1-L to 3-L. Thus, 0, 1 and 3 are the average numerical value of H-bonds observed for 1-L to 3-L, respectively. PVA and lignin are both macromolecular chains. Due to steric hindrance, the PVA chains and lignin molecules do not come close to each other. As a result, the number of H-bonds between them is lower. This reduced H-bonding indicates weaker intermolecular interactions between PVA and lignin, which may

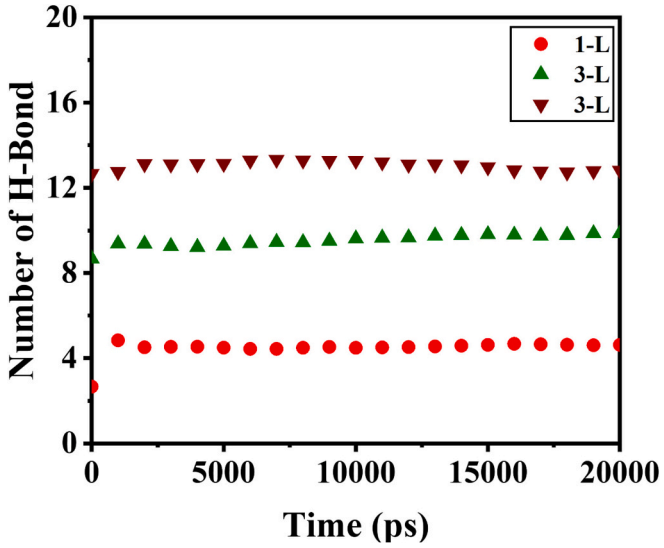


Fig. 4. Number of H-bonds between lignin and lignin molecules.

affect the overall stability and cohesion of the material. Fig. 4 is the representation of the number of H-bond curves for the Lignin-Lignin. The –OH moiety compels lignin to act as an H-bond acceptor as well as a donor [59]. 4, 10 and 13 are the average number of H-bonds of systems 1-L to 3-L, respectively. This increase signifies that as the lignin content rises, the interactions between lignin molecules strengthen. Higher lignin concentrations enhance the formation of H-bonds within the lignin network, leading to greater molecular cohesion and potentially improving the material's structural stability. The number of H-bonds curves are also studied between PVA-PVA and PVA-EG for all systems. It is represented in Fig. 5(a) and Fig. 5(b), respectively. From Fig. 5(a), initially, the H-bonds are increasing, which is due to the structural relaxations in the production run phase. After that, it shows a constant number of H-bonds throughout the dynamics which is given in Table 2 and Fig. S1. This consistency across systems suggests that the presence of lignin does not significantly alter the H-bonding between PVA molecules. It indicates that the PVA-PVA interactions are relatively stable and unaffected by the varying lignin content in these systems. Fig. 5(b) represents the H-bonds between PVA to EG. Initially, the number of H-bonds increases due to structural relaxations during the production run phase. After this initial increase, the number of H-bonds stabilizes and remains constant throughout the dynamics, as shown in Table 2 and Fig. S1. This stability across different systems suggests that lignin does not significantly affect the H-bonding between PVA to EG molecules. It indicates that the interactions between PVA to EG are relatively stable and unaffected by varying lignin content.

3.2. *Structural properties: Radial Distribution Function (RDF) and Coordination Number (CN)*

The molecular dispersion and their interactions are well studied with the help of RDF and *CN*. As the focus of this work to establish the Li^+ ion behaviour in the PVA-eutectogel system, the RDFs are primarily shown with respect to the Li^+ - other components. The RDF is studied between Li^+ ions and different atoms or ions of DES and PVA matrix which are represented in Fig. 6. Fig. 6(a) represents the RDF curve between Li^+ ions and Cl^- ions. For the Li^+ and Cl^- interactions, the first solvation shell is observed at a 3.18 Å distance. The solvation shell refers to the structured arrangement of solvent molecules around a solute particle. The *CNs* are 3.18, 3.28, 3.18, and 3.27 for 0-L, 1-L, 2-L, and 3-L systems, respectively. The first solvation shell is observed due to strong electrostatic interactions between the moieties, which is suggested by Fig. 7. However, the 1-L system has the highest *CN* due to the lignin's direct

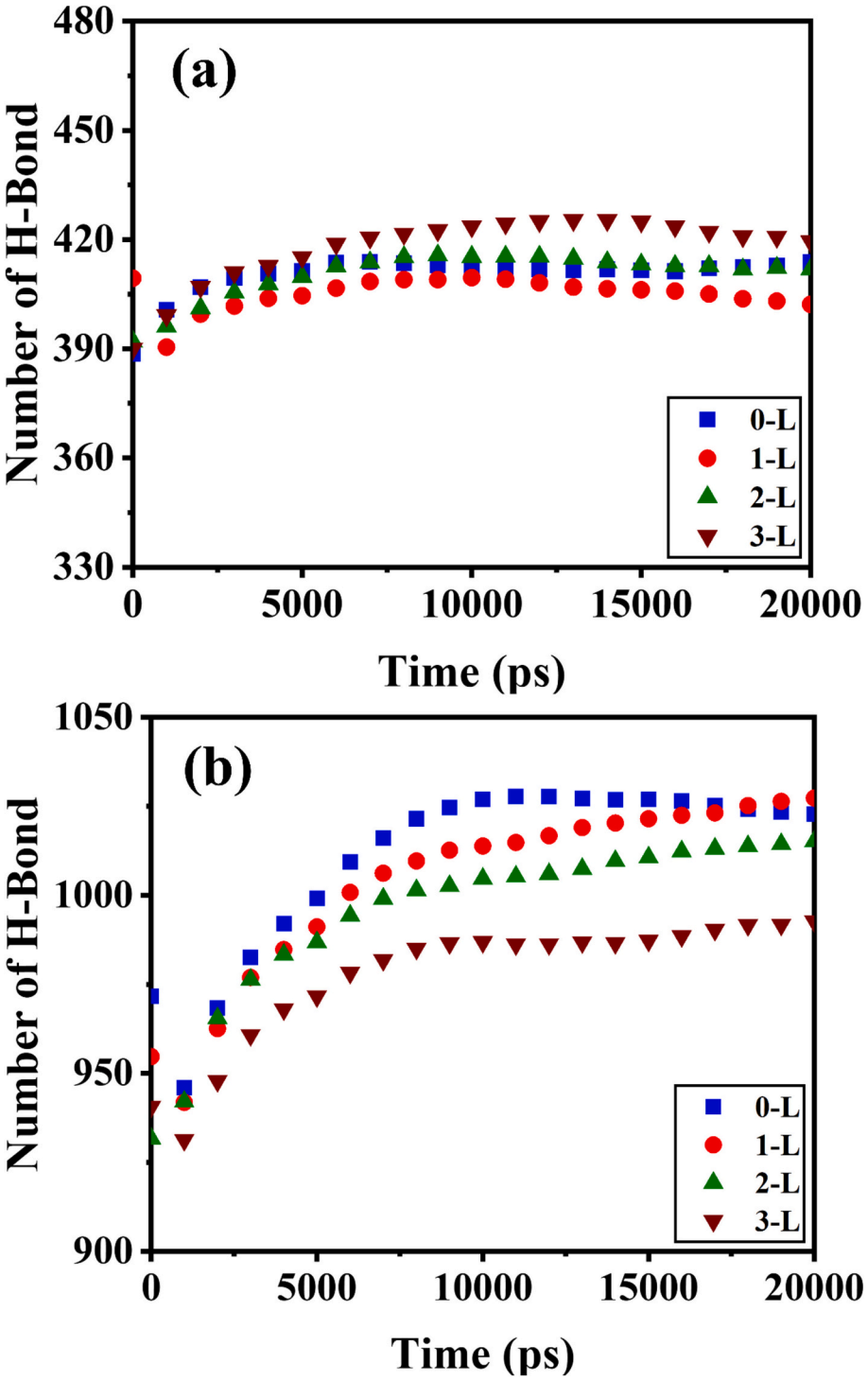


Fig. 5. Number of H-bonds between (a) PVA and PVA molecules (b) PVA and EG.

interaction with the ions. At 6.87 Å, a second solvation shell is observed. The *CN* for 0-L, 1-L, 2-L and 3-L are 16.18, 17.97, 16.17, and 17.7, respectively. The second solvation shell is due to the weak electrostatic interactions. This can be inferred from the Fig. 6(a) that, there is the good amount of ionic interaction between Li^+ and Cl^- ions. Fig. 6(b) shows the RDF curves for Li^+ – EG interactions for all the systems. The first solvation shell is observed at 3.08 Å. The *CNs* are 0.44, 0.35, 0.43, and 0.37 for 0-L, 1-L, 2-L, and 3-L systems, respectively. The first solvation shell is observed due to the electrostatic interactions between the moieties. As lignin introduces the system, the interactions of Li^+ ions with EG are reduced. This is due to the lignin competes with EG for interaction with the Li^+ ions. At 6.58 Å, a second solvation shell is observed. The *CNs* are 7.86, 6.21, 7.69, and 6.58 for 0-L, 1-L, 2-L, and 3-L systems, respectively. After the first solvation shell, the RDF curves are seen to be continuously increasing. However, the RDF values are less than unity. These indicate the formation of small clusters in the system, which is defined by the particles with higher local density than the surrounding medium. The RDF curves for Li^+ – PVA interactions in different systems are represented in Fig. 6(c). The first solvation shell for Li^+ to the oxygen of hydroxyl of PVA interactions is observed at a 3.22 Å

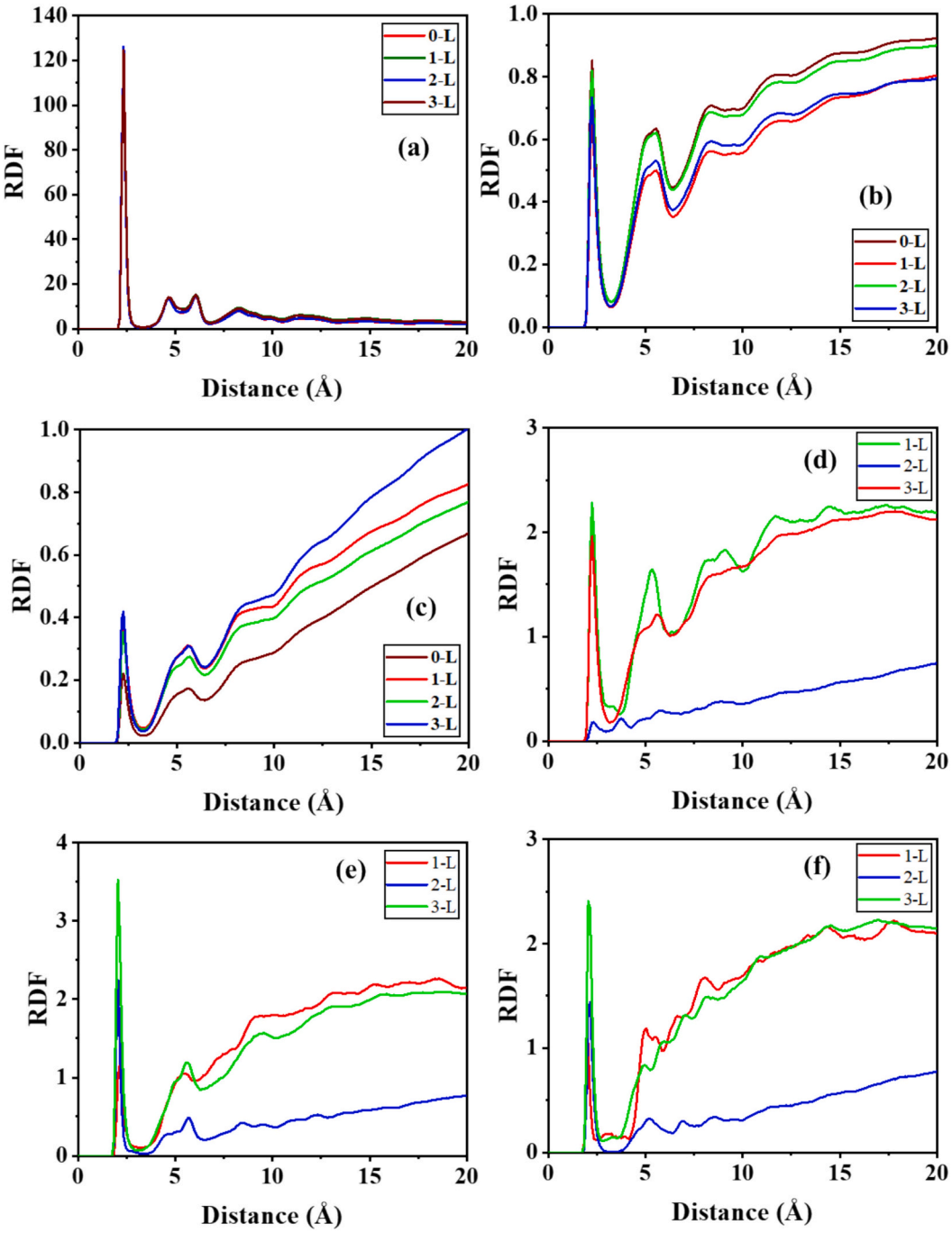


Fig. 6. RDF of (a) Li^+-Cl^- (b) Li^+-EG (c) Li^+-PVA (d) Li^+-OH^- (lignin) (e) Li^+-OCH_3^- (lignin) (f) Li—O (lignin).
I. Schematic representation of Li^+-Cl^- interaction, II. Schematic representation of Li^+-EG interaction, III. Schematic representation of Li^+-PVA interaction, IV. Schematic representation of Li^+-Lignin interaction -OH^- (lignin) interaction (highlighted by orange colour), -OCH_3^- (lignin) interaction (highlighted by lavender colour), —O (lignin) interaction (highlighted by pink colour).

distance. The *CNs* are 0.01, 0.02, 0.02, and 0.02 for 0-L, 1-L, 2-L, and 3-L systems, respectively. The first solvation shell is observed due to the electrostatic interactions between the moieties. The lignin-infused system is showing higher *CN* compared to the 0-L system. This is due to the unfolding of the PVA chain. The steric hindrance introduced by the lignin macromolecules prevents proper folding of the PVA chain. As a results, the Li^+ ions get more binding sites in the PVA chain. Summarising the Fig. 6(a-c), the strongest interactions are showing in the Cl^-. This is due to the nature of ionic interactions. Li^+ and Cl^- are ions with opposite charges, resulting in a strong electrostatic attraction between them.

The RDF between the hydroxy (-OH) moieties of lignin and Li^+ ions is represented in Fig. 6(d). The first solvation shell is spotted at 3.14 Å with a *CN* of $1.00 \times 10^{-3,}$ $3.00 \times 10^{-3,}$ and 2.00×10^{-3} for 1-L, 2-L, and 3-L, respectively. This is due to the weak electrostatic interactions between the moieties. Fig. 6(e) shows the RDF between the the Li^+ – methoxy moiety of lignin. The first solvation shell is observed at 3.34 Å. The *CNs* are 3.30×10^{-4}, 6.67×10^{-4}, and 23.3×10^{-4} for 1-L, 2-L, and 3-L, respectively. This solvation shell is observed due to the electrostatic interactions between the moieties. Fig. 6(f) shows the RDF between the

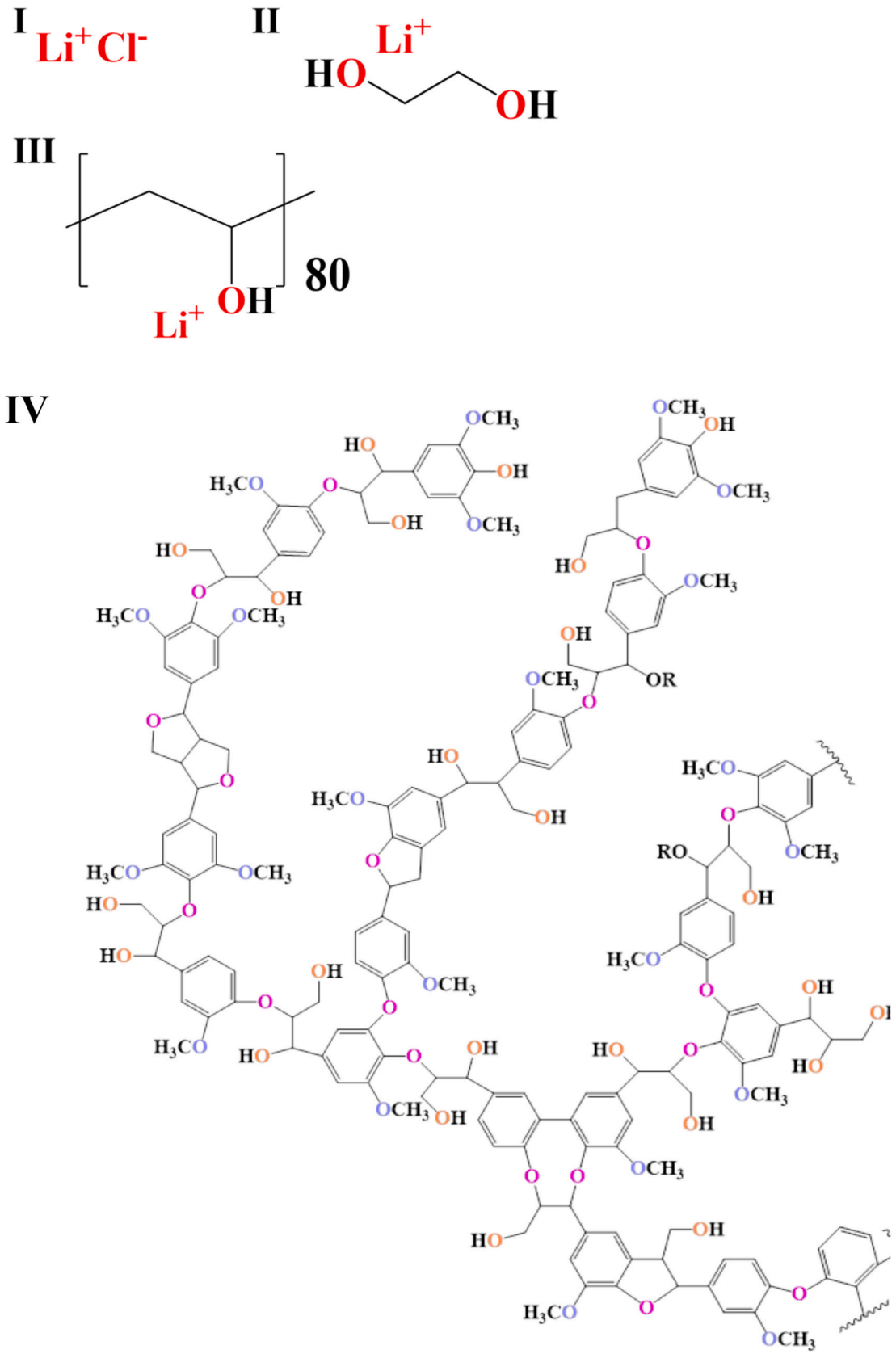


Fig. 6. (*continued*).

lignin (—O—) moiety to the Li^+. The first solvation shell is observed at 3.27 Å. The *CNs* are 1.00×10^{-4}, 6.67×10^{-4}, and 13.3×10^{-4} for 1-L, 2-L, and 3-L, respectively. This is due to the weak electrostatic interactions between the moieties. Comparing the Fig. 6(d-f), the RDF between the methoxy of lignin is higher compares to the other moieties of lignin. This is due to the specific interactions between Li^+ and

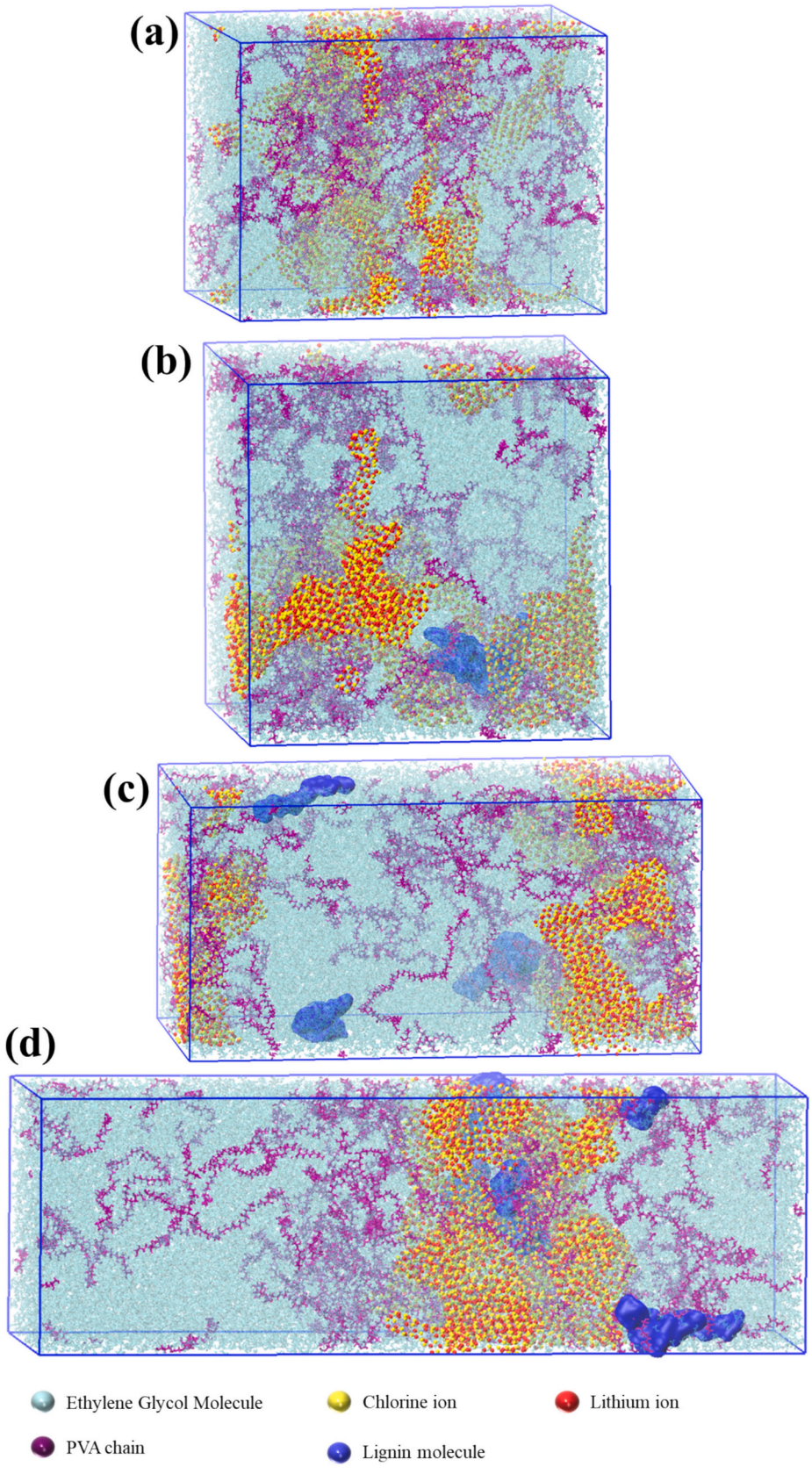


Fig. 7. Simulation snapshot at 20 ns of systems: (a) 0-L (b) 1-L (c) 2-L (d) 3-L.

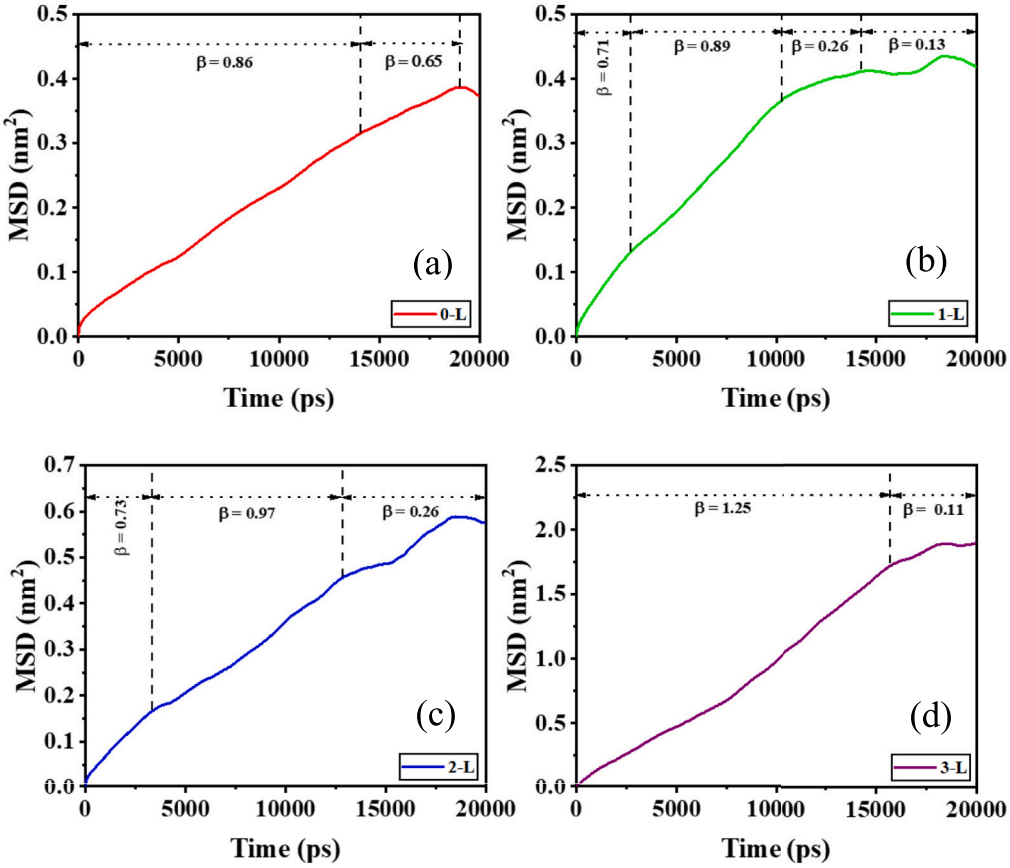


Fig. 8. MSD curve for Li^+ ions present in systems (a) 0-L (b) 1-L (c) 2-L (d) 3-L.

methoxy groups. The methoxy groups (–OCH_3) have a lone pair of electrons on the oxygen atom, which can interact strongly with the positively charged Li^+ ions through electrostatic and possibly Lewis acid-base interactions. The benzene ring of lignin also affects these RDF profiles. The aromatic ring present in lignin creates a unique electronic environment that affects the electron density distribution, which in turn influences the interactions between methoxy groups and lithium ions [60]. The benzene rings may act as electron-rich regions, potentially shielding or competing with methoxy groups for interaction with lithium ions [61]. Additionally, the spatial arrangement of the benzene rings within the lignin structure can create pockets or channels that influence the accessibility of methoxy groups to lithium ions, further affecting the RDF profile. The overall performance of the gel is strongly influenced by the lignin structure, particularly the arrangement and frequency of functional groups such as methoxy and hydroxyl groups on the benzene rings. These groups affect intermolecular interactions, solubility, and crosslinking capabilities. Ether linkages between benzene rings also play a crucial role in determining the flexibility and network formation of the gel. Thus, the specific structure of lignin directly impacts the gel's mechanical properties, swelling behaviour, and ion conductivity, which are essential for its potential applications in various fields.

Fig. S2 depicts the RDF plots for the different moieties present in the lignin to the H moiety of hydroxyl groups of EG. Fig. S2(a) represents the RDF curves for interactions of H moiety of hydroxyl group of EG with ether moieties in lignin (—O—) in 1-L, 2-L and 3-L systems. The first solvation shell is observed at 3.00 Å distance for the interaction between lignin moieties and H moieties of EG due to the weak vdW interactions between the moieties. The second solvation shell is observed at the 5.25 Å due to long-range weak interactions between the moieties. The RDF curve for interactions of H moiety of hydroxyl groups of EG with methoxy moieties (-OCH_3) in lignin is represented by Fig. S2(b). The first solvation shell for this interaction is observed at 2.90 Å. This solvation shell is observed due to the H-bonding interactions between the moieties. Fig. S2(c) display the RDF curves for interactions of the H moiety of hydroxyl groups of EG with the O moiety of hydroxy moieties (-OH) in lignin for different systems. For this interaction, the solvation shell is observed at 5.20 Å distance. This solvation shell is due to the weak electrostatic interactions between the moiety. The hydroxyl moiety of lignin can form H-bonds with the EG solvents. However, it is not forming H-bonds due to the self-assembly of the lignin structure, the hydroxyl moiety of lignin doesn't form H-bonds with the EG.

Fig. S3 is the representation of the RDF plots for the moieties present in the lignin and H moiety of the hydroxyl of PVA. The RDF plot for the ether moieties (—O—) present in the lignin to the H moiety of the hydroxyl of PVA is given in Fig. S3(a). 3.10 Å distance is the distance where the first solvation shell is observed. This solvation shell is observed due to the vdW interactions between the moieties. Fig. S3(b) is the representation of the RDF plot for the methoxy moieties present in the lignin to the H moiety of the hydroxyl of PVA for 1-L, 2-L and 3-L systems, respectively. The first solvation shell is observed at 2.80 Å. This solvation shell is due to the H-bonding interactions between the moieties. The RDF plot for the hydroxy moieties present in the lignin and H moiety of the hydroxyl of PVA for 1-L, 2-L and 3-L systems is given in Fig. S3(c). The first solvation shell for specified interactions is observed at 4.85 Å. This solvation shell is due to the weak electrostatic interactions between the moieties, The hydroxyl moiety of lignin is not forming H-bonds due to the self-assembly of the lignin structure. Summarising Fig. S3 and S4, the methoxy moiety is showing the solvation shell in the H-bonding regions with EG and PVA. The oxygen atom in the methoxy group can

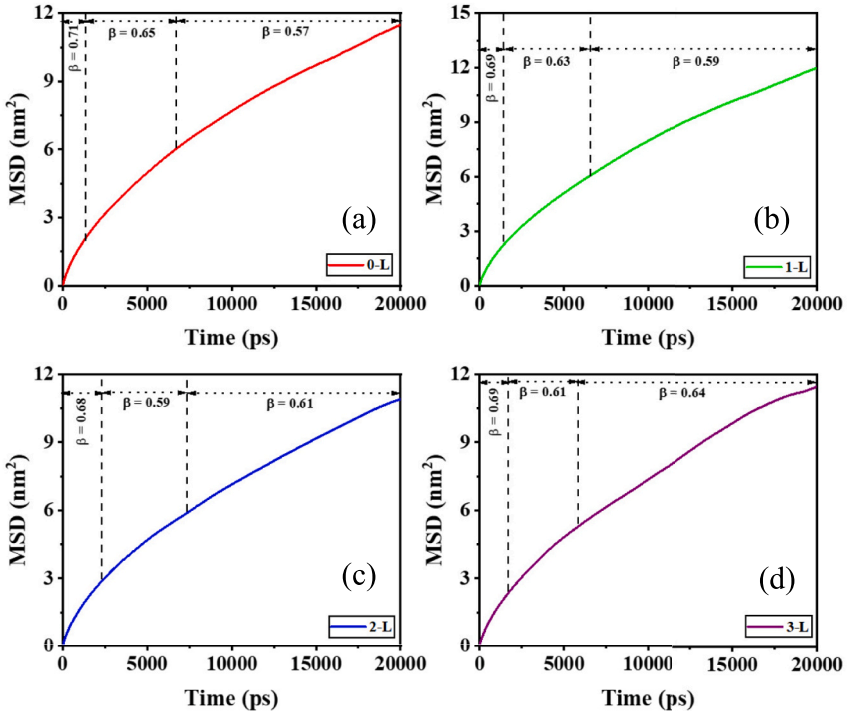


Fig. 9. MSD curve for PVA present in systems (a) 0-L (b) 1-L (c) 2-L (d) 3-L.

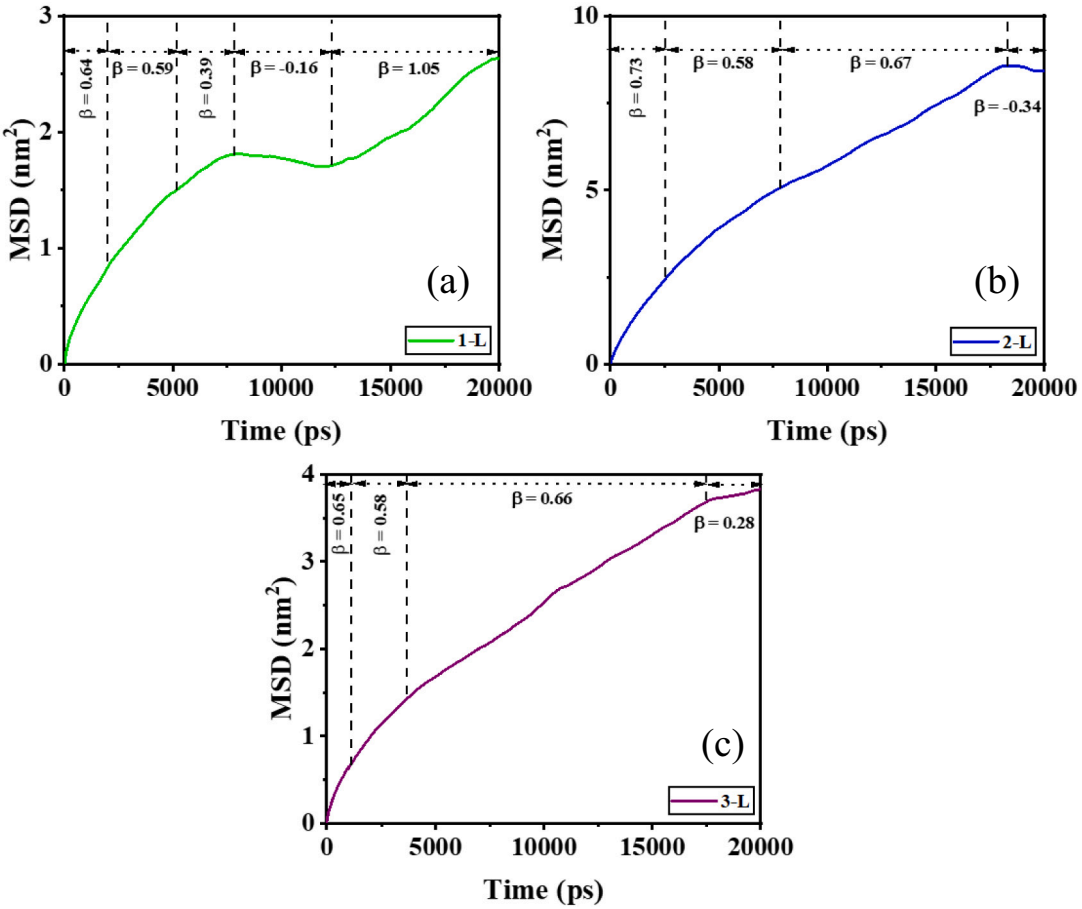


Fig. 10. MSD curve for methoxy moiety of lignin present in systems (a) 1-L (b) 2-L (c) 3-L
Schematic presentation from footnote of Fig. 7.

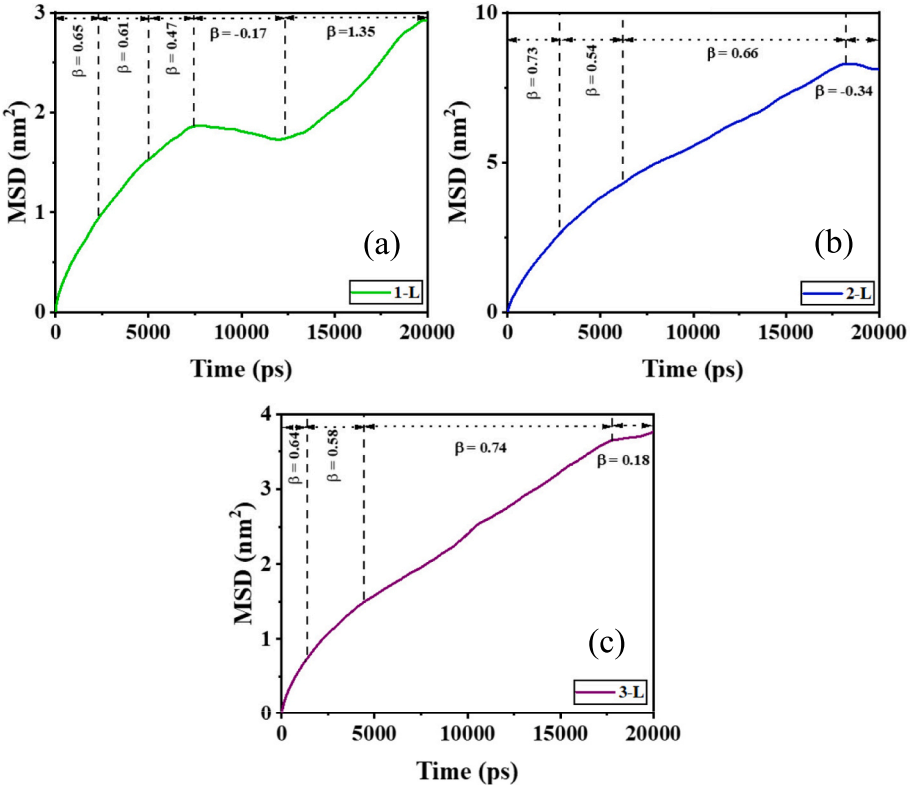


Fig. 11. MSD curve for hydroxyl moiety of lignin present in systems (a) 1-L (b) 2-L (c) 3-L
Schematic presentation from footnote of Fig. 7.

engage in H-bonding more effectively due to its electron density and spatial orientation, facilitating closer interactions. In contrast, the hydroxyl and ether moieties of lignin may not show a clear solvation shell in the H-bonding region due to steric hindrance or less favourable orientation. The hydroxyl group's involvement in intramolecular H-bonding within lignin or other competing interactions could limit its ability to form H-bonds with ethylene glycol or polyvinyl alcohol. Similarly, the ether moiety's geometry may not align well for effective H-bonding, reducing the likelihood of forming a distinct solvation shell.

3.3. Dynamic behaviour of functional moieties through Mean Square Displacement

The MSD plot for Li^+ ions for all the systems (0-L to 3-L) are represented in Fig. 8. The MSD of Li^+ ion for 0-L system is present in the Fig. 8 (a). The overall behaviour of Li^+ ion is sub-diffusive for the 0-L system. The curve is divided in 2 parts as first part is showing β value as 8.60×10^{-1} till 14 ns. This suggests the sub-diffusive behaviour in which Li + ions are stuck in confined space. Further, this value decreases to 6.50×10^{-1} as the H-bond interaction between PVA-PVA and PVA-EG and EG-EG increases with time, which creates a hindrance to the Li^+ ion transportation. Fig. 8(b) shows the MSD plot for Li^+ ion present in the system 1-L. Overall the Li^+ ions show the sub-diffusive behaviour for this system. The MSD curve is divided into 4 parts. For the first part 7.10×10^{-1} is the value for β showing sub-diffusive behaviour. This first part ends at 2.62 ns. The second part ends at 10.240 ns with value of logarithmic slope of MSD plot as 8.90×10^{-1}, representing sub-diffusive behaviour. Between 10.240 ns and 14.280 ns time period β value decreases to 2.60×10^{-1} value. Furthermore, this value decreases to 1.30×10^{-1} till 20 ns. This sub diffusive behaviour observes due to the strong interactions Li^+ and Cl^- ions which suggest in Figs. 6 and 7. As a results, the Li^+ ions stuck higher energy barrier. Due to this barrier the motion of Li^+ ions are restricted. The Li^+ ions' MSD plot for 2-L system is present in the Fig. 8(c) where the plot is divided in 3 parts depending on the value of logarithmic slope of MSD curve. The overall behaviour is seemed to be sub-diffusive. The first part with 7.30×10^{-1} value of β ends at 3.33 ns. This suggests the subdiffusive behaviour, which is due to the restricted motion of Li^+ ions. After that the MSD curve increases till 12.85 ns where β is 9.70×10^{-1}. During this time period, the Li^+ ions attempt to break strong interactions with Cl^-, but the other interacting moieties do not facilitate this process. After that, the MSD is decreases till 20 ns. During this period, the β value is 2.60×10^{-1}.This is come due to the less statistical averaging. The MSD plot for Li^+ ions present in the 3-L system is represented in Fig. 8(d). The MSD curve is divided in 2 parts based on states of diffusion behaviour. For first part the β value is 1.25 showing the diffusion behaviour as super-diffusive behaviour. This super-diffusive behaviour persists till 15.690 ns time. This super diffusive behaviour is shown due to a molecular hopping mechanism, where molecules rapidly move between localized regions within the system. This hopping allows the molecules to overcome energy barriers or escape from potential wells, leading to faster diffusion rates than would be expected under normal diffusion conditions [62]. After that, the value of logarithmic slope of MSD plot significantly decreases to 1.10×10^{-1} which shows the increased resistance to the diffusion of Li^+ ions as number of Lignin molecules increased. As the simulation progresses, the ions interact more with their environment. These interactions create local energy barriers and temporary trapping sites, slowing down the ions' movement. As a result, it shows the sub-diffusive behaviour.

The MSD plots for the PVA moieties are present in Fig. 9. Fig. 9(a-d) is the MSD plot for PVA moieties in the 0-L to 3-L system, respectively. The MSD curve shows the concave downward throughout the dynamics. During the dynamics, the oxygen of hydroxyl moiety of PVA shows subdiffusive behaviour which is suggested by the logarithmic slope of MSD. This subdiffusive behaviour is due to the H-bonding of PVA with

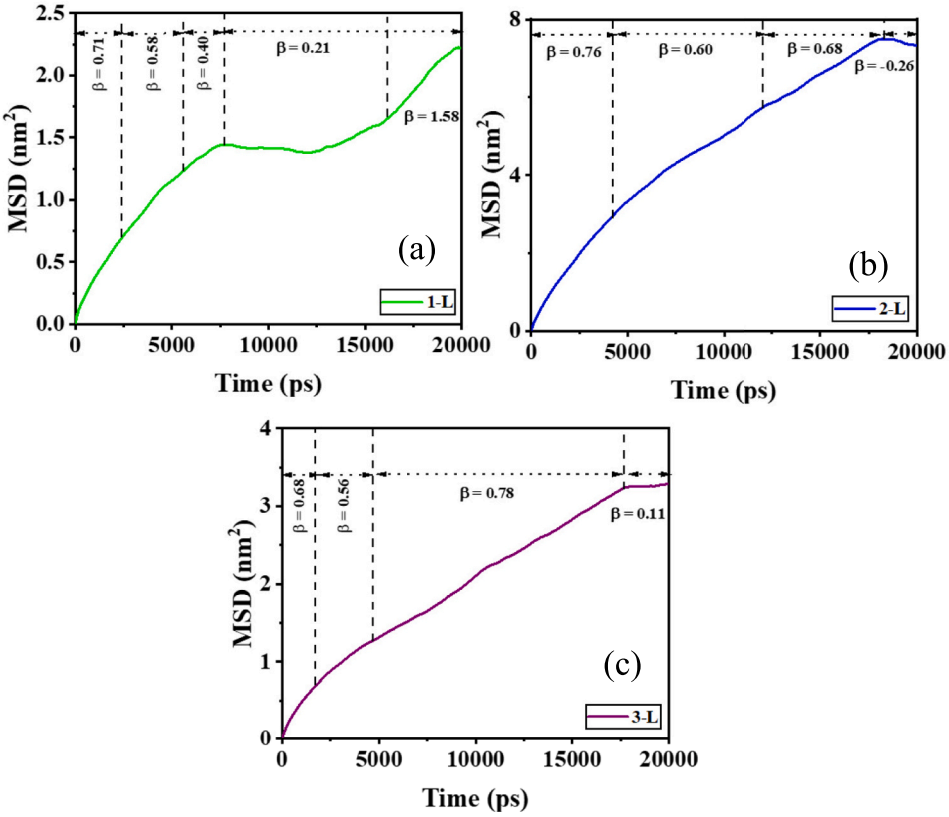


Fig. 12. MSD curve for ether moiety of lignin present in systems (a) 1-L (b) 2-L (c) 3-L.
Schematic presentation from footnote of Fig. 7.

other moieties. This behaviour often signifies the formation of a more stable structure within the system. However, the MSD values for each system are nearly the same. These results are also suggested by H-bonding results. This signifies the lignin doesn't affect the dynamics of PVA.

For understanding the impact of lignin in the gel system, the diffusivity of lignin is crucial which is calculated by the MSD graphs. Fig. 10–12 shows MSD of several moieties present in the lignin system. Fig. 10 represents the MSD plots for methoxy moiety of lignin present in 1-L to 3-L systems, respectively. The MSD plot for methoxy moieties present in 1-L system is shown in Fig. 10(a). Initially, the MSD shows linear behaviour till 7.8 ns. During this time period, the MSD slope changes, but the slope value presents in less than one, which signifies the subdiffusive behaviour. This subdiffusive behaviour is due to its motion being restricted due to its large, complex structure and interactions within the system. After that, the MSD curve shows concave upward till 12.28 ns. During this time periods, the MSD shows subdiffusive behaviour which suggested by the logarithmic slope of MSD. During this time period, the motion of lignin is hindered by the other moieties present in the system and the moiety tries to break the energy barrier for subdiffusive interactions. After that, the MSD curve goes linear end of the dynamics. During this time period, the logarithmic slope of MSD is greater than one, which signifies the superdiffusive behaviour of molecules [62]. This superdiffusive behaviour is due to the molecular hopping mechanism. Fig. 10(b) is the representation of the MSD for methoxy moieties in system 2-L. The plot is divided into 4 parts. The overall behaviour is sub-diffusive. The first part is showing 7.30×10^{-1} β value till 2.48 ns. The β value decreases to 5.80×10^{-1} from 2.48 ns to 7.82 ns time interval. This β value increases to 6.70×10^{-1} till 18.30 ns. During this time period, the MSD shows subdiffusive behaviour and the motion is restricted due to its large, complex structure and interactions within the system. The time interval between 18.30 ns to 20 ns is showing β value -3.40×10^{-1}. This is due to the less statistical averaging of MSD curve. The MSD plot for methoxy moieties present in 3-L system is represented in Fig. 10(c). This plot is divided in 4 parts. The observed β value for the first part is 6.50×10^{-1} showing the sub-diffusive nature of diffusion till 1.05 ns. During this time periods the formation of H-bonds PVA to other moieties and EG to the other moieties restrict the motion of lignin in the solution. It decreases to 5.80×10^{-1} during the time duration 1.05 ns to 3.63 ns. During this time, the decrease of the β value is due to formation of H-bonds of EG and PVA with other moieties, which restricts the motion of lignin. In the third part that is between 3.63 ns to 17.48 ns, the β value increased to 6.60×10^{-1}. During this time periods, the H-bonding results show constant so the MSD results show stable subdiffusive behaviour. This subdiffusive behaviour is due to the interacttions of lignin with other moieties. After this the MSD of lignin moiety is decreased till 20 ns, the β attains 2.80×10^{-1} value. During this time period, the less statistical averaging of MSD curve shows the less β values. Figs. 11 and 12 are similarities found with the Fig. 10.

Fig. S4 is the representation of MSD plots for EG moieties in different systems. Efficient EG diffusion enhances the system's ability to facilitate interactions between macromolecules, supports uniform distribution of additives like lignin, and impacts the material's mechanical and thermal behaviour. The MSD curve shows the linear behaviour. However, the slope of MSD is less than one, which signifies the subdiffusive behaviour. This behaviour likely arises from interactions between EG to the other moieties of components in the system. However, the MSD value of EG nearly constant throughout the all system. This suggests that lignin doesn't affect the diffusion of EG.

4. Conclusion

The working mechanism of Lignin-infused PVA based eutectogel is studied in depth with the help of MD simulations. The dynamics support the observation that, due to the presence of lignin molecule, the characteristics of eutectogel improves. The H-bonding plots reveal that, as the number of lignin molecules increases the H-bonding between Lignin and PVA chains as well as EG molecules strengthens. This suggests the improved strength of eutectogel used as electrolyte in LIBs. From the RDF it can be inferred that, the Li^+ ion transportation is not interrupted by the presence of lignin molecules in the system. The small cluster formation is observed for PVA, EG and lignin molecule. Especially, two lignin molecules present in the system tend to interlock with each other. MSD plots which are helpful to understand the diffusivity of Li^+ ions through the system. Overall, the Li^+ ions show the sub-diffusive nature. Hence, MD simulations provide the deep understanding of the transport as well as structural properties of the eutectogel system. This information might helpful for the further research on eutectogel for different applications.

Ethical approval

Not applicable.

Consent to participate

Not applicable.

Consent to publish

Yes in subscription mode.

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CRediT authorship contribution statement

Aradhana V. Thombre: Writing – original draft, Visualization, Methodology, Investigation, Formal analysis, Data curation. **Rontu Das:** Validation, Methodology, Formal analysis. **Debashis Kundu:** Writing – review & editing, Supervision, Software, Resources, Project administration, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this manuscript.

Data availability

The authors are unable or have chosen not to specify which data has been used.

Appendix A. Supplementary data

The Supplementary Material is about average number of H-bond, Radial Distribution Function (RDF), and Mean Square Displacement (MSD) is included. Supplementary data to this article can be found online at https://doi.org/10.1016/j.est.2024.114060.

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